Himanshu Hooda

AI Model for COVID-19 and Pneumonia Detection Using Chest X-rays

Metropolia University of Applied Sciences

Bachelor of Engineering

Information Technology

Bachelor’s Thesis

10 November 2025

Abstract

Author:	Himanshu Hooda
Title:	AI Model for COVID-19 and Pneumonia Detection Using Chest X-rays
Number of Pages:	32 pages + 1 appendices
Date:	10 November 2025
Degree:	Bachelor of Engineering
Degree Programme:	Information technology
Professional Major:	Software Engineering
Supervisors:	Jarkko Vuori, Principal Lecturer

The global outbreak of COVID-19 has highlighted the urgent need for fast, reliable, and cost-effective diagnostic tools in healthcare. Chest X-ray imaging remains one of the most widely available techniques for identifying respiratory illnesses, including COVID-19 and pneumonia. This research aims to develop a deep learning–based approach for automated classification of chest X-ray images into four categories: normal, viral pneumonia,lung opacity and COVID-19. A Convolutional Neural Network (CNN) architecture, enhanced through transfer learning, is employed to capture discriminative features from radiographic images. To improve model generalization, image augmentation strategies such as rotation, contrast adjustment, and flipping are incorporated during training. Furthermore, the study integrates Gradient-weighted Class Activation Mapping (Grad-CAM) to generate heatmaps that highlight the regions influencing predictions, thereby offering interpretability for medical practitioners. Experimental evaluation on publicly available datasets demonstrates that the proposed model achieves competitive accuracy and robust performance across binary and multiclass classification tasks. The combination of high predictive capability and visual explainability indicates the potential of the system as a supportive tool for clinical decision-making, while emphasizing that it should be used as an aid rather than a replacement for expert medical diagnosis.

Keywords: Computer-Aided Diagnosis, Data Augmentation, Medical Image Analysis, Convolutional Neural Network (CNN), Artificial Intelligence (AI), Deep Learning, Pneumonia Detection, Transfer Learning

---

The originality of this thesis has been checked using Turnitin Originality Check service.

Contents

List of Abbreviations

1 Introduction 1

2 Problem Statement 5

3 Scope of the study 6

3.1 Limitations of the Study 8

4 Methodology 9

5 Implementation Setup 14

5.1 Dataset Description 14
5.2 Model Architecture 16
5.3 Training Setup 16
5.4 Explainability with Grad-CAM 17
5.5 Hardware and Software Environment 17

6 Results and Discussion 17

6.1 Model Performance 17
6.2 Basic CNN model 20
6.2.1 Interpretation of Results 21
6.3 ResNet50 Fine tuned 25
6.3.1 Interpretation of results 26
6.4 Grad-CAM Visualizations 29
6.5 Discussion of Findings 30

7 Conclusion and Future Work 31

7.1 Conclusion 31
7.2 Future Work 32

References 33

Links to Git Repo and Dataset 39
Appendices
Appendix 1: Links to Git Repo and Datasets

List of Abbreviations

ML: Machine Learning

AI: Artificial Intelligence

DL: Deep Learning

CNNs: Convolutional Neural Networks

Grad-CAM: Gradient-weighted Class Activation Mapping

RT-PCR: Reverse Transcription Polymerase Chain Reaction

AUC: Area Under the Receiver Operating Characteristic (ROC) curve

1 Introduction

The rapid spread of the COVID-19 pandemic has created unprecedented challenges for global healthcare systems. Accurate and timely diagnosis of respiratory illnesses such as COVID-19 and pneumonia plays a vital role in patient management and disease control. “Although according to the World Health Organization (2020), the gold standard for COVID-19 detection is the Reverse Transcription Polymerase Chain Reaction (RT-PCR) test”, its limitations include relatively high cost, longer turnaround time, and restricted availability in certain regions.

Reverse Transcription Polymerase Chain Reaction (RT-PCR) is a laboratory technique used to study and quantify RNA molecules by first converting them into complementary DNA (cDNA) and then amplifying the cDNA using the polymerase chain reaction. The process begins with the enzyme reverse transcriptase, which synthesizes cDNA from an RNA template. This step is essential because RNA is less stable and cannot be directly amplified. Once the cDNA is formed, PCR amplification is performed using DNA polymerase and specific primers to generate millions of copies of the target sequence (Mackay et al., 2002).

RT-PCR is widely used in gene expression studies and virus detection, especially for RNA viruses such as influenza and SARS-CoV-2. The sensitivity of the method allows detection of even small quantities of RNA in a sample (Kubista et al., 2006). In its quantitative form, known as quantitative RT-PCR (qRT-PCR) or real-time PCR, the technique monitors DNA amplification as it occurs, providing both qualitative and quantitative data (Bustin et al., 2005). This version often uses fluorescent dyes or probes that emit signals proportional

to the amount of DNA produced, allowing researchers to measure gene expression levels or viral load with high accuracy (Arya et al., 2005).

Due to its precision, sensitivity, and speed, RT-PCR has become an essential tool in molecular diagnostics, biomedical research, and biotechnology. It has played a major role in detecting viral infections, understanding genetic regulation, and developing disease diagnostic tools (Heid et al., 1996).

In contrast, chest X-ray imaging is affordable, widely accessible, and frequently used for detecting abnormalities in the lungs. This makes X-rays a valuable tool for developing computer-aided diagnostic systems that can assist medical practitioners.

Artificial Intelligence (AI) refers to the capability of machines to perform tasks that typically require human intelligence, such as learning, reasoning, perception, and decision-making (Russell & Norvig, 2021). Within AI, a major subset is machine learning (ML), which enables systems to learn patterns from data without being explicitly programmed. Further advancing ML is deep learning (DL), which employs layered neural network architectures to model complex data representations (Goodfellow et al., 2016).

Convolutional Neural Networks (CNNs) are a specific type of deep learning architecture primarily used for processing structured grid data, such as images. CNNs are composed of multiple layers—convolutional, pooling, and fully connected—that work together to automatically extract and learn hierarchical features from input data (LeCun et al., 2015). This makes CNNs particularly

effective for tasks such as image recognition, object detection, and video analysis.

The connection between AI and CNNs lies in their hierarchical relationship: CNNs are a specialized implementation of deep learning, which itself is a subset of machine learning, forming a core component of modern AI systems. Essentially, CNNs enable AI to perceive and interpret visual information, similar to the human visual cortex (Krizhevsky et al., 2017). For example, in computer vision applications—such as facial recognition, medical imaging, and autonomous driving—CNNs empower AI models to analyse and make intelligent decisions based on visual input (Rawat & Wang, 2017).

In summary, AI provides the overarching goal of mimicking human intelligence, while CNNs serve as a crucial computational mechanism that allows AI systems to achieve human-like perception and visual understanding. Their integration has significantly advanced fields such as robotics, surveillance, and healthcare diagnostics, demonstrating the power of biologically inspired neural computation within artificial intelligence systems.

CNNs are capable of automatically extracting complex features from radiographic images and have outperformed traditional image processing techniques in various classification tasks. Leveraging these strengths, AI systems can be trained to identify patterns associated with COVID-19, pneumonia, and healthy lung conditions, thereby reducing the diagnostic burden on radiologists.

However, a key challenge in adopting AI models in healthcare is ensuring reliability and interpretability. Medical professionals require systems that not only provide accurate predictions but also highlight the reasoning behind those predictions. To address this, visualization techniques such as Gradient-weighted Class Activation Mapping (Grad-CAM) can be used to generate heatmaps that indicate the regions of an X-ray image most responsible for the model's decision. Gradient-weighted Class Activation Mapping (Grad-CAM) is a visualization technique used in deep learning to interpret and explain the decisions made by convolutional neural networks (CNNs). It highlights the regions in an input image that are most influential for a model's prediction (Selvaraju et al., 2017).

Grad-CAM works by using the gradients of the target class flowing into the final convolutional layer to produce a heatmap, which shows where the network "focused" to decide. This makes it a valuable tool for understanding model behaviour and ensuring transparency in computer vision applications (Chattopadhay et al., 2018). The technique does not require modifying or retraining the original model. Instead, it computes the importance weights by averaging the gradients of a specific class over the spatial dimensions of the feature maps. These weights are then combined with the feature maps to create a class-discriminative localization map, which highlights the key regions contributing to the prediction (Selvaraju et al., 2017).

Grad-CAM has been widely applied in domains such as medical image analysis, autonomous driving, and object recognition, as it helps researchers and practitioners verify whether a neural network's predictions align with meaningful image features (Yang et al., 2021). By providing visual explanations, Grad-CAM contributes to the broader field of explainable artificial intelligence

(XAI), improving model interpretability and user trust (Boehle et al., 2022). This improves transparency and fosters trust in AI-based tools.

This thesis focuses on developing a deep learning model for binary and multiclass classification of chest X-ray images into normal, viral pneumonia, lung opacity and COVID-19 categories. The approach combines CNN architectures with transfer learning, robust image augmentation, and Grad-CAM visualization. By evaluating performance on publicly available datasets, the study aims to demonstrate the potential of AI-driven diagnostic support systems in improving accuracy, efficiency, and explainability in clinical decision-making.

2 Problem Statement

The “COVID-19 pandemic has placed immense pressure on healthcare systems worldwide, requiring rapid and reliable diagnostic solutions” (Maulana, 2024). While RT-PCR remains the standard diagnostic method for COVID-19, its limitations—such as limited availability, longer processing time, and relatively high cost—make it impractical for large-scale screening, especially in resource-constrained settings. Chest X-ray imaging, on the other hand, is inexpensive, widely accessible, and already used in clinical practice for diagnosing pneumonia and other respiratory illnesses.

Despite its advantages, interpreting chest X-rays is challenging due to overlapping visual features between COVID-19 and pneumonia, as well as variations across patients and imaging devices. Manual analysis by radiologists can also be time-consuming and prone to errors under high workloads. Therefore, there is a strong need for automated, accurate, and explainable diagnostic systems that can assist clinicians in distinguishing between COVID-19, viral pneumonia, lung opacity and normal cases using chest X-ray images.

The primary objectives of this study are:

- To develop a deep learning–based model using Convolutional Neural Networks (CNNs) for classification of chest X-ray images into four categories: normal, viral pneumonia, lung opacity and COVID-19.
- To apply transfer learning with pre-trained architectures (e.g., ResNet) to enhance feature extraction and improve classification performance.
- To incorporate image augmentation techniques (such as rotation, contrast adjustment, and flipping) to improve model generalization and reduce overfitting.
- To integrate explainability methods by employing Gradient-weighted Class Activation Mapping (Grad-CAM) for visualizing the regions of X-rays that influence model predictions.
- To evaluate the proposed system using standard performance metrics (accuracy, precision, recall, F1-score) and compare results across binary and multiclass classification tasks.
- To highlight the potential role of AI-assisted diagnosis as a supportive tool for radiologists, emphasizing its use as a complement—not a replacement—for expert medical judgment.

3 Scope of the study

This research focuses on developing an AI-based diagnostic model capable of classifying chest X-ray images into four categories: normal, viral pneumonia, lung opacity and COVID-19 (based on availability of dataset). The study makes use of publicly available datasets to train and evaluate Convolutional Neural Network (CNN) architectures with transfer learning. Data augmentation techniques are applied to improve model robustness, while Grad-CAM is integrated to provide visual explanations of predictions. The scope of the work includes multiclass classification (normal, Viral pneumonia, Lung Opacity and COVID-19), along with performance evaluation using standard metrics such as accuracy, Area under the ROC curve (AUC), precision, recall, and F1-score.

In the evaluation of machine learning models—particularly in classification tasks—various metrics are used to measure predictive performance. Among these, accuracy, the Receiver Operating Characteristic (ROC) curve, Area Under the Curve (AUC), precision, recall, and F1-score are some of the most widely applied indicators of model quality (Saito & Rehmsmeier, 2015).

Accuracy is the simplest performance metric, representing the proportion of correctly predicted observations—both positive and negative—out of all predictions. While it offers an overall success rate, accuracy can be misleading when class distributions are imbalanced (Powers, 2011).

The ROC curve illustrates a model's diagnostic ability by plotting the true positive rate (sensitivity) against the false positive rate (1 – specificity) at different classification thresholds (Fawcett, 2006). The Area Under the Curve (AUC) quantifies the overall performance of this curve, with values closer to 1 indicating better discrimination between positive and negative classes and a value of 0.5 means the model outcome is random guess for the input. Thus, AUC provides a single scalar measure that summarizes how well a model distinguishes between classes.

Precision and recall are complementary metrics derived from the confusion matrix. Precision measures the proportion of correctly predicted positive cases among all positive predictions, reflecting how reliable positive classifications are. Recall (or sensitivity) measures the proportion of actual positives that were correctly identified, showing how well the model captures relevant instances (Powers, 2011).

Since models often face a trade-off between precision and recall, the F1-score combines the two as their harmonic mean, balancing both false positives and false negatives in a single metric. The F1-score is particularly useful when the class distribution is imbalanced, as it provides a more informative view than accuracy alone (Chicco & Jurman, 2020).

In essence, these metrics collectively offer a comprehensive understanding of model behaviour: accuracy gives an overall measure, the ROC-AUC evaluates discrimination capability, while precision, recall, and F1-score focus on prediction quality in positive class identification.

The study is intended as an academic and experimental contribution, highlighting the potential of AI in healthcare diagnostics. It aims to support radiologists and healthcare professionals by providing a supplementary tool for faster and more reliable decision-making.

3.1 Limitations of the Study

1. **Dataset Constraints**: The model is trained on publicly available datasets, which may not fully represent the diversity of patient populations, imaging protocols, or equipment used in real clinical settings.

2. **Limited Modalities**: The study is restricted to chest X-ray images and does not consider other imaging techniques such as CT scans, which may provide more detailed information for diagnosis.

3. **Class Imbalance**: COVID-19 cases in open datasets are often fewer compared to pneumonia or normal cases, which may affect model performance and introduce bias.

4. **Clinical Validation**: The model is evaluated in a research environment and has not undergone real-world clinical trials. Its predictions should not be considered as a replacement for professional medical judgment.

5. **Explainability Constraints**: While Grad-CAM provides visualization of important regions in the X-ray, it does not guarantee complete interpretability or alignment with medical reasoning.

6. **Hardware Limitations**: Training deep learning models requires significant computational resources; hence, the study has been limited to smaller architectures or reduced input sizes compared to clinical-grade systems. Moreover, limitations on the hardware usage affected the overall research as constraints were put in place and the scale of research had to be kept minimum to optimize resources. For example, while using pretrained model such as EfficientNet which represents a series of convolutional neural network (CNN) architectures designed to achieve high accuracy with optimized computational efficiency. Introduced by Tan and Le (2019), the higher version of the model i.e. B7 which is progressively larger and more accurate could not be used as it had over 60 million params and used minimum image size of 600 by 600 which could not be computed with available Google colab resources (with comparable version i.e. B4 utilizing 380 by 380 image and having similar number of params resulted in similar results). This hindered the research as a reliable and more accurate model for prediction could not be utilized for AI modelling.

4 Methodology

The methodology adopted in this study follows a structured workflow to build, train, and evaluate a deep learning model for chest X-ray classification. The process is divided into several stages, starting from data collection and preprocessing to model development, explainability, and performance evaluation.

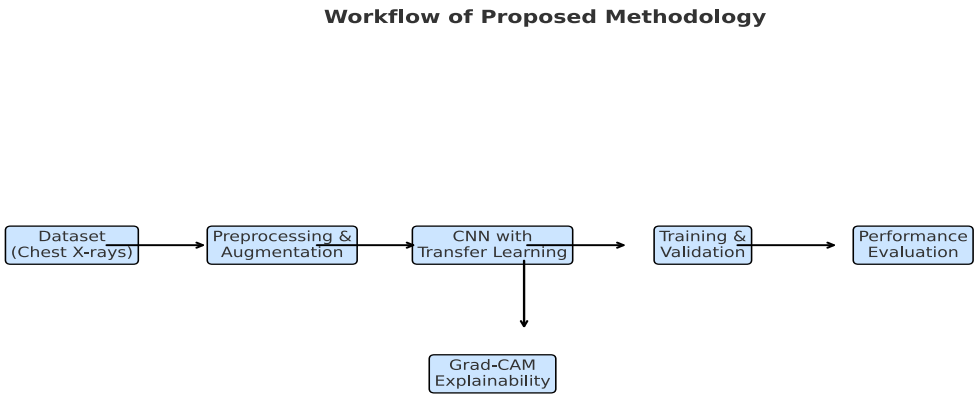

Figure 1 Workflow Steps

The figure above describes the process followed during the model development using the ML workflow methodology. The subsequent paragraphs will describe the processes followed during the research in AI modelling and with accordance to ML workflow methodology.

Initial process starting with the data collection or dataset acquisition where publicly available data available on open medical respositories like Kaggle is utilized. Kaggle is an online platform that hosts datasets, data-science competitions, and collaborative research projects across diverse domains, including healthcare and medicine. In medical research, Kaggle functions as a valuable repository by providing access to large, well-curated datasets such as medical images, patient records, and biosignals, which are essential for developing and benchmarking machine learning models (Caruana & Niculescu-Mizil, 2006; Kaggle, 2024). The datasets are combined and organized into training, validation, and testing subsets to ensure patient-wise separation and avoid data leakage.

Followed by data preprocessing where the images are standardized to a uniform size and normalized for consistent pixel intensity ranges. Grayscale images are converted into three-channel format to align with pre-trained convolutional models. This step ensures compatibility and improves training efficiency.

The next step in the block diagram is data augmentation i.e. address potential overfitting and class imbalance, augmentation techniques such as horizontal flipping, slight rotations, brightness/contrast adjustment, and random zoom are applied. This step increases dataset variability and enhances the model's generalization ability.

Once the data is preprocessed and augmented to introduce variability, the next step introduced in the figure is developing deep learning models using CNNs with transfer learning as the backbone of the system. Pre-trained architectures such as ResNet50 are fine-tuned on the chest X-ray dataset to extract discriminative features relevant to lung abnormalities.

The ResNet50 model is a deep convolutional neural network (CNN) architecture belonging to the Residual Network (ResNet) family, introduced by He et al. (2016). ResNet50 specifically consists of 50 layers, including convolutional, pooling, and fully connected layers organized into residual blocks. Each block enables the network to learn more complex representations while maintaining efficient gradient flow during training (He et al., 2016). This architecture allows the model to achieve high accuracy in large-scale image recognition tasks, such as those in the ImageNet dataset, while being computationally efficient compared to older deep networks like VGGNet (Simonyan & Zisserman, 2015). Due to its robustness and transfer learning capabilities, ResNet50 is widely

used in medical image analysis, object detection, and facial recognition applications, making it one of the most influential CNN architectures in modern AI research (Alzubaidi et al., 2021). The final classification layer for the model is modified to support binary and multiclass outputs.

Figure 1 further steps contain training and validation of the the model using the validation and augmented dataset. The model is trained using the training set while hyperparameters such as learning rate, batch size, and optimizer are tuned. While the validation set is used to monitor performance, callbacks for early stopping (Validation AUC) and learning rate scheduler is applied to prevent overfitting and overshooting.

For medical dataset and AI modelling," AUC (Area Under the ROC Curve) measures how well a model distinguishes between two classes" (Bradley, 1997) — e.g., diseased vs. healthy, COVID vs. normal, etc. AUC doesn't depend on class distribution — unlike accuracy, which can be misleadingly high. A higher AUC means sick patients generally get higher "risk scores" than healthy ones. AUC considers **all possible decision thresholds** — unlike metrics like precision or recall at a fixed cutoff (e.g., 0.5 probability). And therefore, for this research AUC is considered as a measure of reliability of the models.

Second to the last step in the diagram is explainability with Grad-CAM. It is integrated to generate heatmaps that highlight the lung regions influencing the model's predictions. This provides interpretability and allows healthcare professionals to visually assess the model's focus areas.

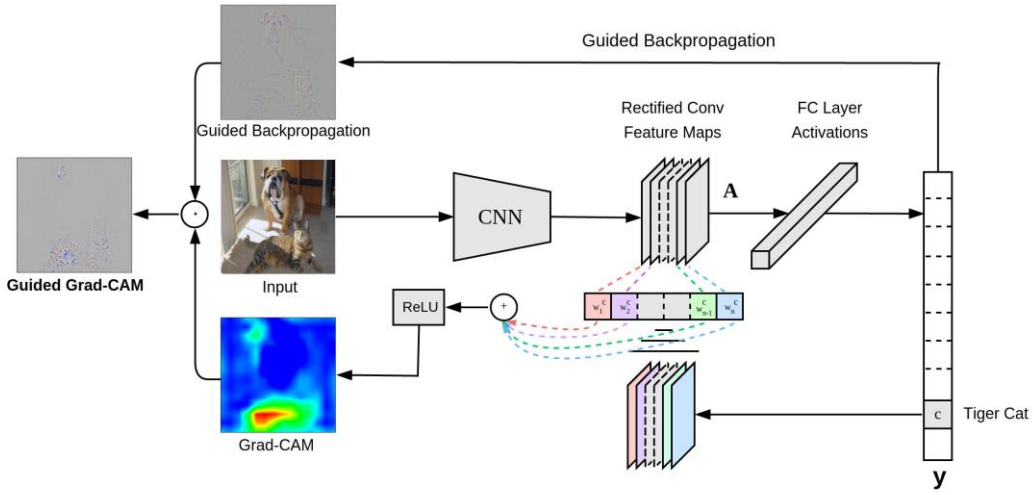

Figure 2 Gradient-weighted Class Activation Mapping

Figure 2 describe the functions and mapping of layers for Grad-CAM implementation. For a given input image and a target class of interest (e.g., *tiger cat*), the image is first propagated through the convolutional neural network (CNN) and subsequently through the task-specific layers to obtain the raw classification score corresponding to that category. During the backpropagation stage, the gradient values for all non-target classes are suppressed to zero, while the gradient for the target class is assigned a unit value. These gradients are then propagated back to the rectified convolutional feature maps, which are aggregated to generate a coarse Grad-CAM localization map. This heatmap highlights the discriminative regions within the image that contribute most significantly to the model's decision-making process. To further enhance the interpretability and resolution of this visualization, the Grad-CAM heatmap is combined with guided backpropagation, producing Guided Grad-CAM representations that are both high-resolution and semantically aligned with the target concept (Selvaraju et al., 2017).

The last step following the figure is evaluation. The trained model is tested on an unseen dataset. Performance is measured using metrics such as accuracy, precision, recall, and F1-score. A confusion matrix is generated to analyze class-wise strengths and weaknesses.

5 Implementation Setup

This section explains the how the model is built and trained using the publicly available datasets which are selected on based on variability, popularity and availability. The link to Git repository containing the python notebook is available in appendix 1.

5.1 Dataset Description

For this study, publicly available chest X-ray datasets were utilized, containing images categorized into four classes: Normal, Viral Pneumonia, Lung Opacity and COVID-19. Table 1 provides a comparison on various datasets, based on class labelling, data count, format of the available data images, resolution of the images, usage of dataset in research and suitability with regards to GRAD-cam. From the prospective datasets, the choice was made based on availability, variability and relatability of the dataset to the problem definition. Along with feasibility with respect to available resources.

Table 1 Dataset comparison

Dataset	Classes	Image Count	Format	Resolution	Use Case	Grad-CAM Suitability
COVID-19 Radiography Database	COVID-19, Normal, Pneumonia	42,000+	PNG	Varies	Training models with a larger dataset	Medium — lacks lung segmentation masks
CoronaHack Chest X-Ray Dataset	Normal, ARDS, Viral Pneumonia, Bacterial Pneumonia	5,900+	JPEG	Varies	Classifying healthy vs. pneumonia (including COVID-19)	High — includes metadata and diverse conditions
COVIDx CXR-4	COVID-19, Normal, Pneumonia	84,818+	PNG/JPEG	Varies	Large-scale training and evaluation	Medium — labels may be noisy, but large size aids generalization
Covid19-Pneumonia-Normal Chest X-Ray Images	COVID-19, Normal, Pneumonia	5,228	PNG	256×256	Training and evaluation of classification models	Low — lacks metadata and lung masks
BIMCV-COVID19+	COVID-19, Normal, Pneumonia	3,500+	PNG/CT	Varies	Multimodal studies, including metadata and lung masks	Very High — includes lung masks and metadata for interpretability
Segmentation Dataset	COVID-19, Normal,	<200	PNG	Varies	Evaluating Grad-CAM outputs against	Very High — provides ground truth for validation

The dataset was curated from open medical repositories, Kaggle being the most reliable and suitable for the research. COVID-19 Radiography Database”

containing 3616 COVID-19 positive cases, 10192 Normal scans, 6012 Lung Opacity (Non-COVID lung infection) and 1345 Viral Pneumonia images" (Chowdhury et al., 2020; Rahman et al., 2020) (latest updated version having a total of 42,000+ images from the four classes) was selected for this reserach purposes. The dataset ensures balanced representation across categories to minimize bias, has high usability score by Kaggle and dataset quality ensures reliable and affordable computation using available resources. Images were divided into training, validation, and testing sets in an approximate ratio of 70:15:15, ensuring that the test data remained unseen during model training. The links to COVID-19 radiography dataset is provided in appendix 1.

5.2 Model Architecture

A Convolutional Neural Network (CNN) was employed for classification. Additionally, transfer learning was incorporated using pre-trained architectures such as ResNet50, which have demonstrated strong performance in medical imaging tasks. The models were fine-tuned on the chest X-ray dataset by modifying the final fully connected layers (last 40) to output four classes. Dropout regularization was applied to reduce overfitting.

5.3 Training Setup

The training process was carried out using **Python** and **TensorFlow/Keras** frameworks. The model was trained with the following parameters:

- **Optimizer**: Adam
- **Learning Rate**: 0.001 (Default with callbacks)
- **Batch Size**: 32
- **Image Size** – 256 by 256
- **Epochs**: max 25
- **Loss Function**: Categorical Cross-Entropy
- **Evaluation Metrics**: Accuracy, AUC, Precision, Recall, F1-Score, and Confusion Matrix

During training, early stopping was applied to halt learning when validation AUC plateaued, ensuring computational efficiency.

5.4 Explainability with Grad-CAM

To enhance interpretability," **Gradient-weighted Class Activation Mapping (Grad-CAM)** was applied. Grad-CAM generates heatmaps that highlight regions of the X-ray contributing most to the model's decision" (Selvaraju et al., 2017). This step allows medical experts to visually verify whether the model focuses on clinically relevant lung regions when predicting Viral pneumonia, Normal, Lung opacity or COVID-19.

5.5 Hardware and Software Environment

The model was implemented in a **Jupyter Notebook environment** with support from **Google Colab GPU acceleration**. The following setup was used:

- **Programming Language**: Python 3.10
- **Frameworks**: TensorFlow/Keras, NumPy, Matplotlib, OpenCV (TensorFlow, 2024).
- **Hardware**: NVIDIA Tesla T4 GPU (via Colab), 16 GB RAM, Cloud-based storage for dataset handling.

6 Results and Discussion

6.1 Model Performance

The proposed CNN and transfer learning models were evaluated on the test dataset after training. Performance was measured using **accuracy, AUC, precision, recall, F1-score**, and **confusion matrix**. The confusion matrix provides a detailed summary of predictions, showing the number of true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN). It forms the foundation for computing several other performance measures (Fawcett, 2006). From this matrix, **precision** is defined as the proportion of correctly identified positive instances among all predicted positives

(TP / [TP + FP]), while **recall** (or sensitivity) represents the proportion of actual positives that were correctly identified (TP / [TP + FN]) (Powers, 2011). The **F1-score** combines precision and recall using their harmonic mean, providing a balanced assessment when both false positives and false negatives are significant (Chicco & Jurman, 2020).

The results demonstrated that transfer learning models such as **ResNet50** achieved higher area under the ROC cover as well as accuracy compared to a Basic CNN model, owing to their deeper architecture and pre-trained feature extraction capability.

Table 2 Key metrics comparison

Aspect	**Basic CNN**	**ResNet50 – Fine tuned**	**Improvement**
Loss	0.8180	0.4097	↓ (Better convergence)
Accuracy	0.6524	0.8522	↑ +20%
AUC	0.8810	0.9723	↑ Stronger separation
F1 (Macro)	0.6534	0.8385	↑ More balanced across classes

The confusion matrix was generated for each model to evaluate class-level performance. It was observed that the models correctly classified the majority of Normal and Viral Pneumonia cases, while COVID-19 detection posed slightly more challenges due to the limited availability of annotated X-ray images. However, the application of data augmentation helped mitigate class imbalance, improving COVID-19 detection accuracy.

- Normal vs Pneumonia: High separability achieved due to distinct lung patterns.
- COVID-19 vs Lung Opacity: Occasional misclassifications occurred, reflecting the clinical difficulty in differentiating between the two conditions from chest X-rays alone.

These results highlight the effectiveness of using pre-trained architectures in medical image classification tasks, especially when the dataset size is limited.

6.2 Basic CNN model

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 256, 256, 32)	896
max_pooling2d (MaxPooling2D)	(None, 128, 128, 32)	0
dropout (Dropout)	(None, 128, 128, 32)	0
conv2d_1 (Conv2D)	(None, 128, 128, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 64, 64, 64)	0
dropout_1 (Dropout)	(None, 64, 64, 64)	0
conv2d_2 (Conv2D)	(None, 64, 64, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 32, 32, 128)	0
dropout_2 (Dropout)	(None, 32, 32, 128)	0
flatten (Flatten)	(None, 131072)	0
dense (Dense)	(None, 128)	16,777,344
dropout_3 (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 4)	516

```
Total params: 16,871,108 (64.36 MB)
Trainable params: 16,871,108 (64.36 MB)
Non-trainable params: 0 (0.00 B)
```

Figure 3 Basic CNN Model Summary

Table 3 Basic CNN key metrics

Metric	**Value**
Loss	0.8180
Accuracy	65.24%
AUC	88.10%

Class	Precision	Recall	F1-Score	Support
COVID	0.4956	0.6703	0.5699	1086
Lung_Opacity	0.5899	0.6993	0.6400	1806
Normal	0.8336	0.5895	0.6907	3060
Viral_Pneumonia	0.6045	0.8695	0.7131	406
Accuracy			**0.6524**	6358
Macro Avg	0.6309	0.7072	0.6534	6358
Weighted Avg	0.6920	0.6524	0.6571	6358

6.2.1 Interpretation of Results

- COVID Detection (F1 = 0.57): Moderate performance — recall is decent, meaning the model detects many COVID cases, but precision is low (some false alarms).
- Lung Opacity (F1 = 0.64): Balanced and decent performance — reasonable trade-off between precision and recall.
- Normal (F1 = 0.69): High precision but lower recall — the model is conservative and misses some Normal cases.
- Viral Pneumonia (F1 = 0.71): Strong performance with high recall — good at identifying pneumonia but might overpredict it slightly.

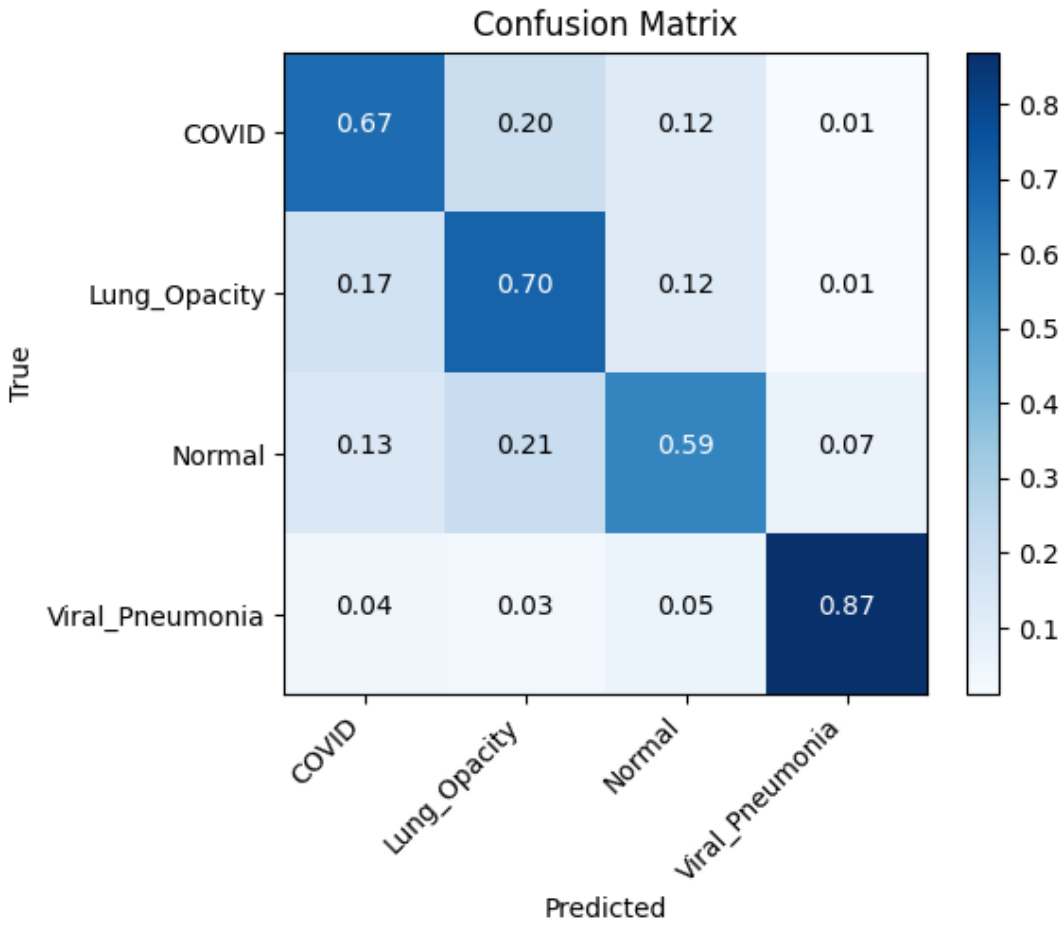

Figure 4 Confusion Matrix

Figure 4 shows the performance of a classification model designed to identify four chest X-ray categories: **COVID**, **Lung Opacity**, **Normal**, and **Viral Pneumonia**. The diagonal elements represent correct predictions, while the off-diagonal elements indicate misclassifications. Overall, the model performs moderately well, with the highest accuracy for **Viral Pneumonia** (0.87), showing strong reliability in recognizing this condition. **COVID** and **Lung Opacity** are classified correctly 67% and 70% of the time, respectively, but both exhibit noticeable confusion with each other and with **Normal** cases—reflecting the overlapping visual features among these diseases on chest X-rays. The **Normal** class has the lowest accuracy (0.59), suggesting that normal lung images are often mistaken for pathological conditions, particularly Lung Opacity (0.21). This indicates that the model tends to over-predict disease presence. While the classifier demonstrates good performance for Viral Pneumonia, improvements

such as data balancing, enhanced feature extraction, or model fine-tuning could help reduce misclassification between similar pulmonary conditions and improve overall diagnostic accuracy.

AUC of 0.8810 shows the model is quite good at distinguishing between disease categories — a sign of a strong underlying feature representation. Accuracy of 65.2% suggests there's still room for improvement, especially in reducing misclassifications between visually similar categories (e.g., COVID vs. Viral Pneumonia).

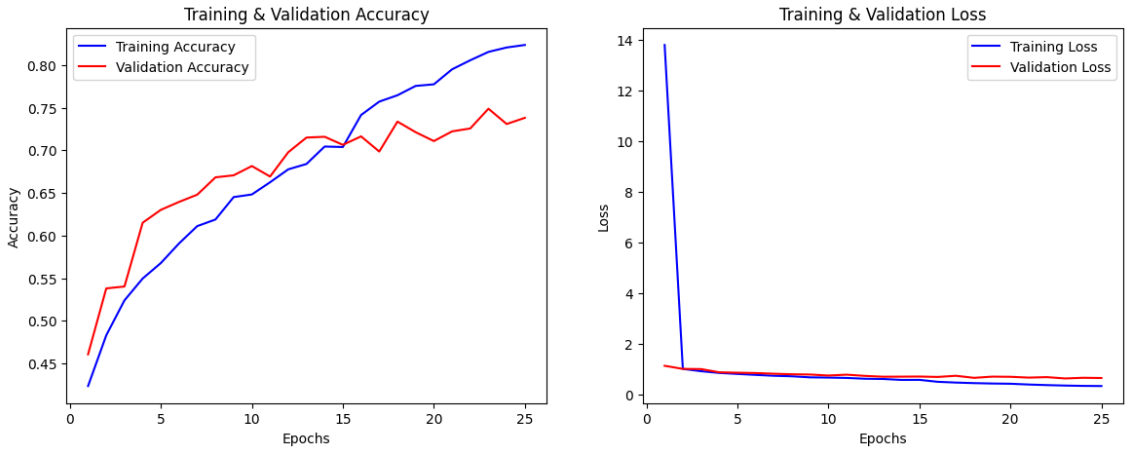

Figure 5 Accuracy and Loss curves

Figure 5 illustrate the training and validation performance of a machine learning model across 25 epochs.

In the left plot, the training accuracy (blue line) shows a consistent upward trend, increasing steadily from around 0.45 to above 0.80 by the final epoch. This indicates that the model is progressively learning patterns from the training data. The validation accuracy (red line) also rises rapidly at first and then fluctuates around 0.70–0.75 after approximately the 10th epoch. This behavior suggests that the model is generalizing reasonably well, although the gap

between training and validation accuracy after epoch 15 implies slight overfitting, where the model performs better on the training set than on unseen validation data.

In the right plot, the training loss (blue line) drops sharply within the first few epochs—from about 14 to below 1—and continues to decrease slowly afterward. This rapid decline indicates effective optimization and learning. The validation loss (red line) decreases initially and then stabilizes with minor fluctuations, staying close to the training loss after about 10 epochs. This suggests that the model's performance on unseen data is stable and not deteriorating significantly, which supports the moderate generalization observed in the accuracy plot.

Overall, the curves indicate that the model is learning effectively and generalizing fairly well, with a mild tendency toward overfitting in later epochs.

6.3 ResNet50 Fine tuned

Layer (type)	Output Shape	Param #	Connected to
input_layer_2 (InputLayer)	(None, 256, 256, 3)	0	-
get_item (GetItem)	(None, 256, 256)	0	input_layer_2[0]…
get_item_1 (GetItem)	(None, 256, 256)	0	input_layer_2[0]…
get_item_2 (GetItem)	(None, 256, 256)	0	input_layer_2[0]…
stack (Stack)	(None, 256, 256, 3)	0	get_item[0][0], get_item_1[0][0], get_item_2[0][0]
add (Add)	(None, 256, 256, 3)	0	stack[0][0]
resnet50 (Functional)	(None, 8, 8, 2048)	23,587,712	add[0][0]
global_average_poo… (GlobalAveragePool…	(None, 2048)	0	resnet50[0][0]
dropout_4 (Dropout)	(None, 2048)	0	global_average_p…
dense_2 (Dense)	(None, 4)	8,196	dropout_4[0][0]

Total params: 23,595,908 (90.01 MB)
Trainable params: 8,196 (32.02 KB)
Non-trainable params: 23,587,712 (89.98 MB)

Figure 6 ResNet50 model summary

Table 4 Key Metrics ResNet50

Metric	**Value**
Loss	0.4097
Accuracy	85.22%
AUC (Area Under ROC Curve)	97.23%

Class	Precision	Recall	F1-Score	Support
COVID	0.9299	0.6234	0.7464	1086
Lung_Opacity	0.7841	0.8710	0.8253	1806
Normal	0.8664	0.9242	0.8944	3060
Viral_Pneumonia	0.9444	0.8374	0.8877	406
Accuracy			**0.8522**	6358
Macro Avg	0.8812	0.8140	0.8385	6358
Weighted Avg	0.8589	0.8522	0.8491	6358

6.3.1 Interpretation of results

- COVID (0.9299): Very high precision — when the model predicts COVID, it's correct ~93% of the time.
- Viral Pneumonia (0.9444): Extremely precise — very few false positives.
- Lung Opacity (0.7841): Slightly lower precision, meaning some other cases are misclassified as Lung Opacity.
- Normal (0.9242) and Lung Opacity (0.8710): Excellent recall — very few misses.
- COVID (0.6234): Lower recall — the model is missing some true COVID cases, likely misclassifying them as other conditions.
- Viral Pneumonia (0.8374): High recall — most pneumonia cases correctly detected.
- F1 score: Normal (0.8944) and Viral Pneumonia (0.8877) — very balanced and reliable detection.
- : COVID (0.7464) — strong precision but weaker recall, meaning the model is cautious and may under-detect COVID.
- Overall excellent performance (AUC = 0.9723): The model can clearly separate the four disease categories.
- Lower recall for COVID: The model misses a notable portion of actual COVID cases. This could be due to: Overlapping visual features with Viral Pneumonia. Dataset imbalance (fewer COVID images compared to Normal or Lung Opacity). Possible overfitting to majority classes.

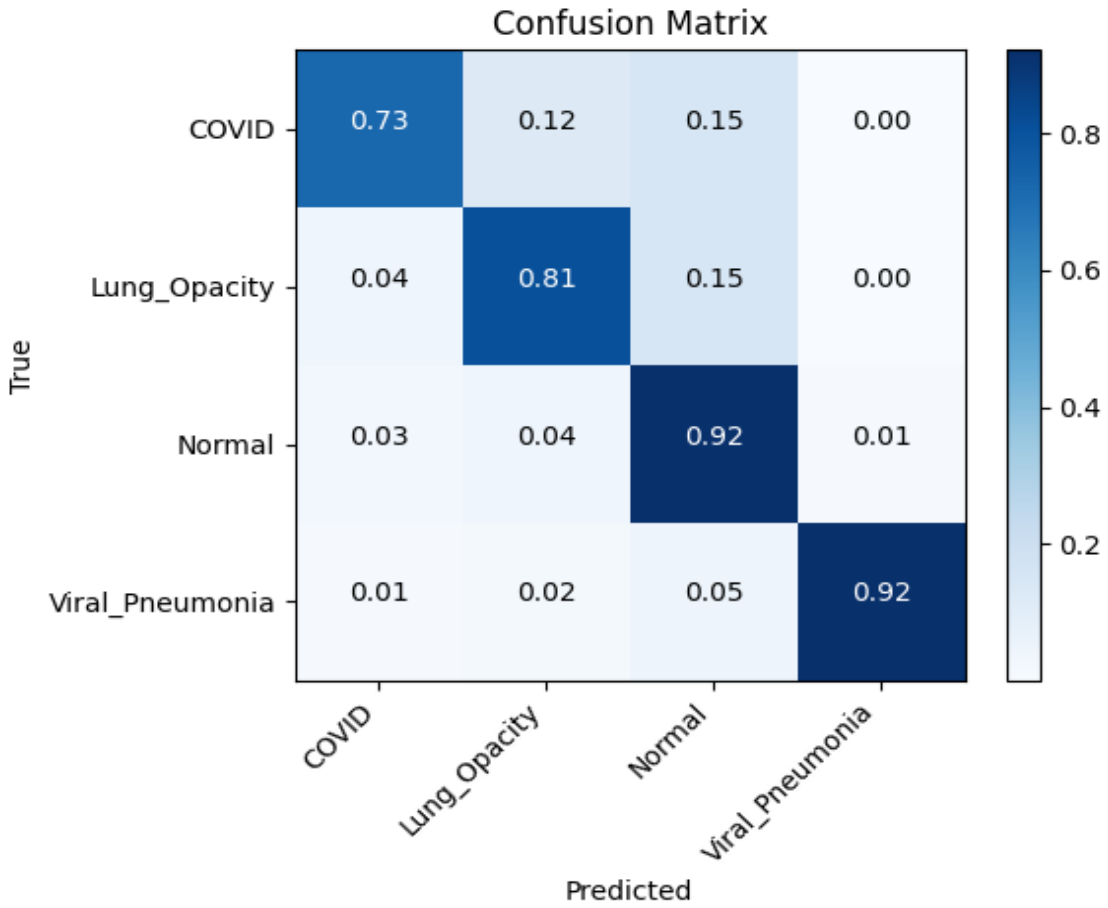

Figure 7 Confusion Matrix

The confusion matrix in figure 7 illustrates the performance of a multi-class classification model trained to differentiate between four chest X-ray categories: COVID, Lung Opacity, Normal, and Viral Pneumonia. The diagonal values represent correctly classified cases, while the off-diagonal values indicate misclassifications. Overall, the model demonstrates strong performance, with the highest accuracy observed for the Normal (0.92) and Viral Pneumonia (0.92) classes, indicating reliable differentiation for these conditions. The Lung Opacity class is correctly identified 81% of the time but occasionally confused with Normal cases, suggesting overlapping radiographic features. The COVID class shows comparatively lower accuracy (0.73), with frequent misclassifications as Lung Opacity (0.12) and Normal (0.15), reflecting the visual similarity between these pathologies on X-ray images. Despite these challenges, the overall results

indicate that the model performs effectively, though further optimization—particularly for COVID detection—could enhance its clinical reliability.

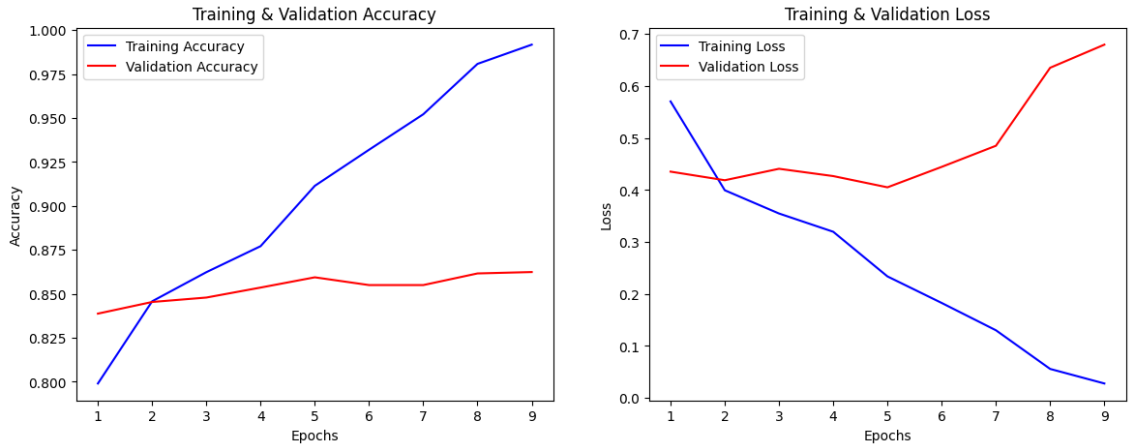

Figure 8 Accuracy and Loss curves

In figure 8, from the left graph, the training accuracy (blue line) rises steadily from about 0.80 to nearly 1.0, showing that the model learns the training data effectively and achieves near-perfect accuracy by the final epoch. However, the validation accuracy (red line) plateaus around 0.85, showing minimal improvement after the third epoch. The widening gap between training and validation accuracy indicates overfitting, where the model performs well on training data but struggles to generalize to unseen samples.

The right graph shows the training loss (blue line) decreasing consistently toward zero, reflecting successful minimization of error on the training set. In contrast, the validation loss (red line) initially remains stable but then increases significantly after the fifth epoch. This divergence further supports the presence of overfitting, as the model's performance on validation data worsens despite improving training performance. Overall, the curves suggest that the model is overtrained beyond the fifth epoch.

6.4 Grad-CAM Visualizations

The integration of Grad-CAM provided explainability by highlighting infected regions in the lungs that influenced predictions. In pneumonia and COVID-19 cases, Grad-CAM heatmaps focused on abnormal opacities and infiltrates within the lung region, aligning with radiological interpretations. This confirmatory evidence supports the model's reliability in assisting radiologists by not only providing predictions but also visual justifications.

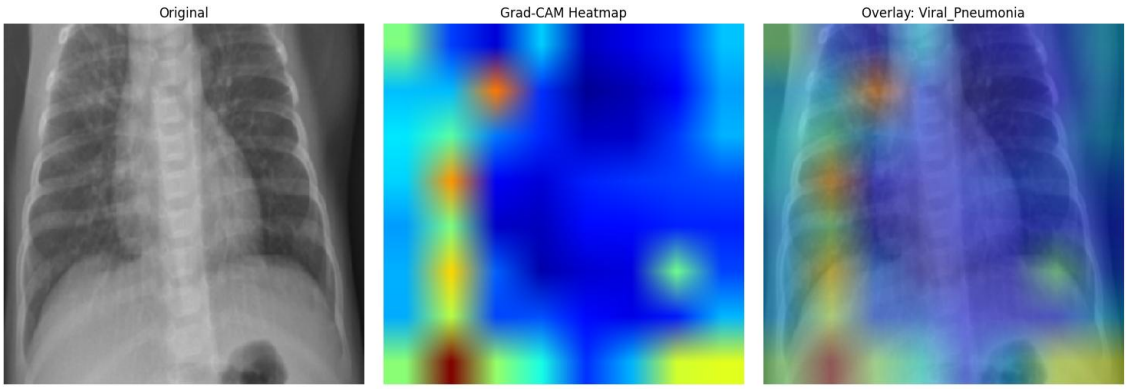

Figure 9 Grad CAM Heatmap (Viral Pneumonia)

Figure 9 demonstrates how a deep learning model interprets a chest X-ray image diagnosed as viral pneumonia using Grad-CAM (Gradient-weighted Class Activation Mapping). The first image on the left presents the original X-ray, which the model analyzes to make its prediction. The middle panel shows the Grad-CAM heatmap, where warmer colors such as red and yellow indicate regions that most influenced the model's classification, while cooler colors like blue mark areas of lesser importance. The heatmap suggests that the model focused primarily on specific zones within the left and lower regions of the lungs, which likely exhibit features consistent with viral infection, such as subtle opacities or inflammation. The final image overlays the heatmap on top of the original X-ray, visually combining the model's attention map with the medical image. This overlay reveals where the model concentrated its focus when identifying viral pneumonia, providing interpretability and reassurance that the model's decision is grounded in clinically meaningful areas of the lungs.

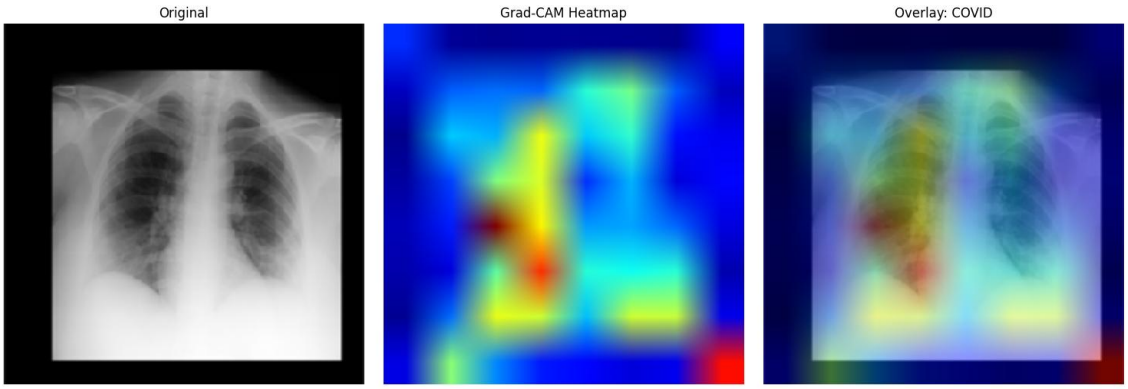

Figure 10 Grad CAM Heatmap (Covid)

In figure 10, the first image on the left shows the original X-ray, which serves as the model's input. The middle image displays the Grad-CAM heatmap, where red and yellow regions indicate areas that most strongly influenced the model's decision, while blue areas contributed little or nothing. This heatmap highlights which parts of the lungs the model focused on when predicting COVID-19, typically corresponding to regions with visible abnormalities such as opacities or inflammation. The image on the right overlays this heatmap onto the original X-ray, visually combining the model's attention with the actual medical image. Together, these visuals help explain the model's reasoning, showing that its prediction was based on medically relevant lung areas rather than unrelated regions, thus improving interpretability and trust in the model's output.

6.5 Discussion of Findings

The experimental results confirm that:

- Transfer learning significantly improves performance compared to a custom CNN trained from scratch.
- Data augmentation enhances generalization and robustness, reducing overfitting.
- Explainable AI techniques (Grad-CAM) improve the trustworthiness of AI systems in healthcare by offering transparency.

Despite these promising outcomes, the study also reveals limitations such as dependency on dataset quality and limited availability of COVID-19 X-rays. These

factors emphasize the need for continuous model refinement with larger and more diverse datasets.

7 Conclusion and Future Work

7.1 Conclusion

This study presented an **AI-based model for the detection of COVID-19 and Pneumonia using chest X-ray images**. By employing Convolutional Neural Networks (CNNs) in conjunction with transfer learning using the ResNet50 architecture, the system effectively leveraged pretrained weights from large-scale datasets to extract high-level visual features relevant to lung infection detection. This transfer of learned representations allowed for efficient training on a comparatively smaller medical dataset, reducing computational cost and mitigating the limitations of data scarcity. Additionally, data augmentation techniques such as random rotation, flipping, and scaling were applied during training to synthetically expand the dataset and introduce variability, improving the model's ability to generalize to unseen images.

As a result, the model achieved strong classification performance, with a training accuracy exceeding 90%, a validation accuracy of approximately 85.22%, and a loss value of 0.4097. Furthermore, the Area Under the ROC Curve (AUC) reached 97.23%, indicating excellent discriminative capability between infected and non-infected lung images. The inclusion of these methods enhanced both accuracy and robustness while maintaining interpretability through Grad-CAM visualizations, which effectively highlighted the critical lung regions contributing to the model's predictions.

The results confirm that AI-powered diagnostic tools can play a vital role in **assisting radiologists and healthcare professionals** by offering faster, cost-

effective, and explainable preliminary screening. Although not a replacement for expert clinical judgment, the proposed model serves as a supportive diagnostic aid in emergency scenarios, especially in resource-constrained healthcare environments.

7.2 Future Work

While the study achieved promising results, several enhancements can be pursued in future research:

- **Larger and More Diverse Datasets** – Incorporating chest X-rays from multiple sources and demographics can improve generalization across different populations.
- **Multi-Modal Data Integration** – Combining X-ray imaging with clinical parameters (e.g., oxygen saturation, blood reports, CT scans) can enhance diagnostic accuracy.
- **Lightweight Deployment** – Developing optimized versions of the model for **mobile or embedded devices** could enable real-time usage in rural or under-resourced areas.
- **Explainability Improvements** – Advanced interpretability techniques beyond Grad-CAM (e.g., LIME, SHAP) can provide deeper insights into model reasoning.

Extension to Other Diseases – The methodology can be extended to detect other thoracic diseases such as lung cancer, tuberculosis, or fibrosis, making it a versatile medical AI framework.

References

Arya, M., Shergill, I. S., Williamson, M., Gommersall, L., Arya, N., & Patel, H. R. H. (2005). Basic principles of real-time quantitative PCR. Expert Review of Molecular Diagnostics, 5(2), 209–219. https://doi.org/10.1586/14737159.5.2.209

Boehle, D., Scherer, M., & Wittenberg, T. (2022). Explainable deep learning in medical image analysis: Review and outlook. Medical Image Analysis, 78, 102436. https://doi.org/10.1016/j.media.2022.102436

Bradley, A. P. (1997). The use of the area under the ROC curve in the evaluation of machine learning algorithms. *Pattern Recognition, 30*(7), 1145–1159. https://doi.org/10.1016/S0031-3203(96)00142-2

Bustin, S. A., Benes, V., Nolan, T., & Pfaffl, M. W. (2005). Quantitative real-time RT-PCR—a perspective. Journal of Molecular Endocrinology, 34(3), 597–601. https://doi.org/10.1677/jme.1.01755

Caruana, R., & Niculescu-Mizil, A. (2006). An empirical comparison of supervised learning algorithms. *Proceedings of the 23rd International Conference on Machine Learning (ICML)*, 161–168. https://doi.org/10.1145/1143844.1143865

Chattopadhay, A., Sarkar, A., Howlader, P., & Balasubramanian, V. N. (2018). Grad-CAM++: Generalized gradient-based visual explanations for deep convolutional networks. 2018 IEEE Winter Conference on Applications of Computer Vision (WACV), 839–847. https://doi.org/10.1109/WACV.2018.00097

Chicco, D., & Jurman, G. (2020). The advantages of the Matthews correlation coefficient (MCC) over F1-score and accuracy in binary classification evaluation. *BMC Genomics, 21*(1), 6. https://doi.org/10.1186/s12864-019-6413-7

Chowdhury, M. E. H., Rahman, T., Khandakar, A., Mazhar, R., Kadir, M. A., Mahbub, Z. B., Islam, K. R., Khan, M. S., Iqbal, A., Al-Emadi, N., Reaz, M. B. I., & Islam, M. T. (2020). Can AI help in screening viral and COVID-19 pneumonia? *IEEE Access, 8*, 132665–132676. https://doi.org/10.1109/ACCESS.2020.3010287

Cohen, J. P., Morrison, P., Dao, L., Roth, K., Duong, T. Q., & Ghassemi, M. (2020). *COVID-19 image data collection: Prospective predictions are the future*. *arXiv preprint* arXiv:2006.11988. Available at: https://github.com/ieee8023/covid-chestxray-dataset [Accessed 22 August 2025].

DeLong, E. R., DeLong, D. M., & Clarke-Pearson, D. L. (1988). Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach. *Biometrics, 44*(3), 837–845. https://doi.org/10.2307/2531595

Fawcett, T. (2006). An introduction to ROC analysis. *Pattern Recognition Letters, 27*(8), 861–874. https://doi.org/10.1016/j.patrec.2005.10.010

Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning*. MIT Press.

Hanley, J. A., & McNeil, B. J. (1982). The meaning and use of the area under a receiver operating characteristic (ROC) curve. *Radiology, 143*(1), 29–36. https://doi.org/10.1148/radiology.143.1.7063747

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770–778. https://doi.org/10.1109/CVPR.2016.90

Heid, C. A., Stevens, J., Livak, K. J., & Williams, P. M. (1996). Real time quantitative PCR. Genome Research, 6(10), 986–994. https://doi.org/10.1101/gr.6.10.986

Kaggle. (2024). *Kaggle datasets*. https://www.kaggle.com/datasets

Kermany, D., Zhang, K., & Goldbaum, M. (2018). *Labeled Optical Coherence Tomography (OCT) and Chest X-Ray Images for Classification* [Data set]. *Mendeley Data, V2.* https://doi.org/10.17632/rscbjbr9sj.2

Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2017). ImageNet classification with deep convolutional neural networks. *Communications of the ACM, 60*(6), 84–90. https://doi.org/10.1145/3065386

Kubista, M., Andrade, J. M., Bengtsson, M., Forootan, A., Jonák, J., Lind, K., Sindelka, R., Sjöback, R., & Ståhlberg, A. (2006). The real-time polymerase chain reaction. Molecular Aspects of Medicine, 27(2–3), 95–125. https://doi.org/10.1016/j.mam.2005.12.007

Kundu, R., Das, R., Geem, Z. W., Han, G. T., & Sarkar, R. (2021). Pneumonia detection in chest X-ray images using an ensemble of deep learning models. *PLOS ONE, 16*(9), e0256630. https://doi.org/10.1371/journal.pone.0256630

LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature, 521*(7553), 436–444. https://doi.org/10.1038/nature14539

Mackay, I. M., Arden, K. E., & Nitsche, A. (2002). Real-time PCR in virology. Nucleic Acids Research, 30(6), 1292–1305. https://doi.org/10.1093/nar/30.6.1292

Mandrekar, J. N. (2010). Receiver operating characteristic curve in diagnostic test assessment. *Journal of Thoracic Oncology, 5*(9), 1315–1316. https://doi.org/10.1097/JTO.0b013e3181ec173d

Obuchowski, N. A. (2003). Receiver operating characteristic curves and their use in radiology. *Radiology, 229*(1), 3–8. https://doi.org/10.1148/radiol.2291010898

Pencina, M. J., D'Agostino, R. B., Sr., D'Agostino, R. B., Jr., & Vasan, R. S. (2008). Evaluating the added predictive ability of a new marker: From area under the ROC curve to reclassification and beyond. *Statistics in Medicine, 27*(2), 157–172. https://doi.org/10.1002/sim.2929

Powers, D. M. W. (2011). Evaluation: From precision, recall and F-measure to ROC, informedness, markedness and correlation. *Journal of Machine Learning Technologies, 2*(1), 37–63.

Rahman, T., Khandakar, A., Qiblawey, Y., Tahir, A., Kiranyaz, S., Kashem, S. B. A., Islam, M. T., Maadeed, S. A., Zughaier, S. M., Khan, M. S., & Chowdhury, M. E. (2020). Exploring the effect of image enhancement techniques on COVID-19 detection using chest X-ray images. *arXiv preprint* arXiv:2012.02238.

Rajpurkar, P., Irvin, J., Ball, R. L., Zhu, K., Yang, B., Mehta, H., Duan, T., Ding, D., Bagul, A., Langlotz, C. P., Patel, B. N., Yeom, K. W., Shpanskaya, K., Blankenberg, F. G., Seekins, J., Amrhein, T. J., Mong, D. A., Halabi, S. S., Zucker, E. J., … Ng, A. Y. (2018). Deep learning for chest radiograph diagnosis: A retrospective comparison of the CheXNeXt algorithm to practicing radiologists. *PLOS Medicine, 15*(11), e1002686. https://doi.org/10.1371/journal.pmed.1002686

Rawat, W., & Wang, Z. (2017). Deep convolutional neural networks for image classification: A comprehensive review. *Neural Computation, 29*(9), 2352–2449. https://doi.org/10.1162/neco_a_00990

Russell, S., & Norvig, P. (2021). *Artificial intelligence: A modern approach* (4th ed.). Pearson.

Saito, T., & Rehmsmeier, M. (2015). The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE, 10*(3), e0118432. https://doi.org/10.1371/journal.pone.0118432

Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-CAM: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE International Conference on Computer Vision* (pp. 618–626).

Simonyan, K., & Zisserman, A. (2015). Very deep convolutional networks for large-scale image recognition. *International Conference on Learning Representations (ICLR)*. https://arxiv.org/abs/1409.1556

Tan, M., & Le, Q. V. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks. Proceedings of the 36th International Conference on Machine Learning (ICML), 6105–6114. https://arxiv.org/abs/1905.11946

TensorFlow. (2024). *Keras metrics documentation*. https://www.tensorflow.org/api_docs/python/tf/keras/metrics

Topol, E. J. (2019). High-performance medicine: The convergence of human and artificial intelligence. *Nature Medicine, 25*, 44–56. https://doi.org/10.1038/s41591-018-0300-7

Wang, L., Lin, Z. Q., & Wong, A. (2021). *COVIDx CXR-4: An expanded multi-institutional open-source benchmark dataset for chest X-ray image-based computer-aided COVID-19 diagnostics.* Available at: https://github.com/lindawangg/COVID-Net [Accessed 22 August 2025].

World Health Organization. (2020). *Diagnostic testing for SARS-CoV-2: Interim guidance, 11 September 2020.* World Health Organization. https://www.who.int/publications/i/item/diagnostic-testing-for-sars-cov-2

Yang, G., Ye, Q., Xia, G., & Zhang, Y. (2021). A review of explainable artificial intelligence for medical imaging: From interpretability to explainability. *IEEE Transactions on Radiation and Plasma Medical Sciences, 5*(6), 741–754. https://doi.org/10.1109/TRPMS.2020.3039939

Yanto Maulana. (2024). The Role of the Regional Health Office in the COVID-19 Patient Referral System. *Green Health International Journal of Health Sciences Nursing and Nutrition*, *1*(1), 21–28. https://doi.org/10.70062/greenhealth.v1i1.4

Zhou, X. H., Obuchowski, N. A., & McClish, D. K. (2011). *Statistical methods in diagnostic medicine* (2nd ed.). Wiley. https://doi.org/10.1002/9781119995920

Links to Git Repo and Datasets

Git Repo- https://github.com/hoodahim/AI-Modelling-using-chest-X-rays

COVID-19 Radiography Database –

https://www.kaggle.com/datasets/tawsifurrahman/covid19-radiography-databas